

Семинар 4

24 февраля 2026 г. 17:15

Исходные

Решение

оч. 1.
Равно
 $\alpha(T=20^\circ\text{C}) = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P = 3 \cdot 10^{-4} \text{K}^{-1}$
 $K = -V \left(\frac{\partial P}{\partial V} \right)_T = 2,2 \cdot 10^9 \text{Па}$
 $\Delta T = 1^\circ\text{C}$
 $\Delta P_{\text{исх}} = ?$

Используем правило диф-я
 $\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$
 $\left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial P}{\partial V} \right)_T \cdot \left(\frac{\partial V}{\partial T} \right)_P = K \cdot \alpha$
 $\Delta P = K \cdot \alpha \cdot \Delta T =$ ищем
 $= 2,2 \cdot 10^9 \cdot 3 \cdot 10^{-4} \cdot 1 = 6,6 \cdot 10^5 \text{Па}$

$\left(\frac{\partial u}{\partial V} \right)_T = 0$ $P = T f(V)$
 $du = Tds - pdv$
 $\left(\frac{\partial u}{\partial V} \right)_T = T \left(\frac{\partial s}{\partial V} \right)_T - P$

оч. 2 $P = T f(V)$
о. 3. $\left(\frac{\partial u}{\partial V} \right)_T = 0$

Расширение Вальтера

Термодинамические преобразования.

$du_{(s,v)} = Tds - pdv$

$d(u - Ts) = dF_{(T,v)} = -s dT - p dv$

$d(u + pV) = dH = Tds + Vdp$

$d(u + pV - Ts) = dG = -s dT + V dp$

$\left(\frac{\partial u}{\partial s} \right)_v \left(\frac{\partial u}{\partial v} \right)_s = \frac{\partial^2 u}{\partial s \partial v} = \frac{\partial^2 u}{\partial v \partial s}$

$\left(\frac{\partial T}{\partial v} \right)_s = - \left(\frac{\partial p}{\partial s} \right)_T ; \left(\frac{\partial s}{\partial v} \right)_T = \left(\frac{\partial p}{\partial T} \right)_v$

$\left(\frac{\partial T}{\partial p} \right)_s = \left(\frac{\partial v}{\partial s} \right)_p$

$- \left(\frac{\partial s}{\partial p} \right)_T = \left(\frac{\partial v}{\partial T} \right)_p$

Получим на ур-е состояния (универсальная)

$dZ(x, y) = \left(\frac{\partial Z}{\partial x} \right)_y dx + \left(\frac{\partial Z}{\partial y} \right)_x dy = 0$

пусть Z константа

$\left(\frac{\partial Z}{\partial x} \right)_y \left(\frac{\partial y}{\partial Z} \right)_x = - \left(\frac{\partial Z}{\partial y} \right)_x$

$\left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial Z} \right)_z \left(\frac{\partial Z}{\partial x} \right)_y = -1$

Единицы

№ 5.16

Решение:

$T_0 = 273 \text{K}$
 $P_0 = 100000 \text{Па}$
 V_0
 $\left. \begin{matrix} T_0 = 273 \text{K} \\ P_0 = 100000 \text{Па} \\ V_0 \end{matrix} \right\} \Rightarrow \left. \begin{matrix} \delta Q = 0 \\ T = ? \\ P = 100000 \text{Па} \\ V > V_0 \end{matrix} \right\}$

$\alpha = \frac{1}{V_0} \left(\frac{\partial V}{\partial T} \right)_P =$ $\left(\frac{\partial T}{\partial P} \right)_s = ?$

$\tilde{c}_P = 0,033$

$\gamma =$

$\tilde{c}_P = \frac{T}{m} \left(\frac{\partial s}{\partial T} \right)_P$

$\left(\frac{\partial T}{\partial P} \right)_s \left(\frac{\partial s}{\partial T} \right)_P \left(\frac{\partial P}{\partial s} \right)_T = -1$

$= - \left(\frac{\partial T}{\partial V} \right)_P$

$\left(\frac{\partial T}{\partial P} \right)_s \left(\frac{\partial s}{\partial T} \right)_P \left(\frac{\partial T}{\partial V} \right)_P = 1$

№ 5.28

$\rho, \mu, \gamma = \frac{c_P}{c_V} = \frac{\left(\frac{\partial s}{\partial T} \right)_P}{\left(\frac{\partial s}{\partial T} \right)_V}$

$\left(\frac{\partial P}{\partial T} \right)_V = ?$ $\alpha_T = \frac{1}{V_0} = \left(\frac{\partial V}{\partial T} \right)_P, \beta_T = \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$

$\left(\frac{\partial P}{\partial T} \right)_V = \frac{V_0}{\beta_T} = +V_0 \alpha_P$

$\left(\frac{\partial P}{\partial T} \right)_V = \frac{\alpha_P}{\beta_T}$

$ds(T, V) = c_V \frac{dT}{T} + \left(\frac{\partial s}{\partial V} \right)_T (dV)_s = 0$

$ds(T, P) = c_P \frac{dT}{T} + \left(\frac{\partial s}{\partial P} \right)_T dP = 0$

$\frac{c_P}{c_V} = \frac{\beta_T}{\beta_S}$